

Symmetric polynomials and their relation to conformal field theory

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Outline

- 1 What is a conformal field theory?
- 2 Conformal field theory from representation theory
- 3 Symmetric polynomials and their applications

Why do we care about conformal field theories?

Conformal field theories are quantum field theories that are symmetric with respect to conformal transformations, transformations that preserve angles but not necessarily lengths. They describe

- string theory from the world sheet perspective,
- statistical systems at critical points.

Conformal field theories are also interesting because

- they are exactly solvable toy models for particle physics,
- they have beautiful mathematical structure.

What is a conformal field theory?

A (2 dimensional) conformal field theory essentially consists of two things:

- 1 A vector space $\mathcal{H}$ called the space of states.
- 2 An assignment $v^{(1)}, \dots, v^{(n)} \mapsto f(z_1, \dots, z_n; v^{(1)}, \dots, v^{(n)})$, for $n \geq 1$, where $v^{(i)} \in \mathcal{H}$, $z_i \in \mathbb{P} = \mathbb{C} \cup \{\infty\}$ and f is real smooth with no singularities away from diagonals $z_i = z_j, i \neq j$.

The functions $f(z_1, \dots, z_n; v^{(1)}, \dots, v^{(n)})$ are called n -point correlation functions and are usually denoted as

$$f(z_1, \dots, z_n; v^{(1)}, \dots, v^{(n)}) = \langle v^{(1)}(z_1) \cdots v^{(n)}(z_n) \rangle.$$

The correlation functions correlate the system being in various states at various points in space-time and thus encode all physical quantities that can in principle be measured.

What is a conformal field theory?

Since the theory is supposed to be symmetric with respect to conformal transformations (transformations that preserve angles but not lengths), the correlation functions must transform “nicely” with respect to said transformations.

In 2 dimensions the Lie algebra of infinitesimal conformal transformations is infinite-dimensional. Thus, correlation functions must satisfy infinitely many constraints. This allows many conformal field theories to be solved exactly from symmetry considerations alone.

What is a conformal field theory?

Conformal symmetry guarantees the existence of two subspaces $V, \bar{V} \subset \mathcal{H}$, such that for any $v \in V$, $\bar{v} \in \bar{V}$ and any $w^{(1)}, \dots, w^{(n)} \in \mathcal{H}$

$\langle v(z) w^{(1)}(\zeta_1) \cdots w^{(n)}(\zeta_n) \rangle$ is meromorphic in z (possible poles at ζ_i),
 $\langle \bar{v}(z) w^{(1)}(\zeta_1) \cdots w^{(n)}(\zeta_n) \rangle$ is anti-meromorphic in z (possible poles at ζ_i).

Thus, we can Laurent expand $v(z) = \sum_{m \in \mathbb{Z}} v_m z^{-m-1}$

$$\begin{aligned} \langle v_m w^{(1)}(\zeta_1) \cdots w^{(n)}(\zeta_n) \rangle &= \oint_C z^m \langle v(z) w^{(1)}(\zeta_1) \cdots w^{(n)}(\zeta_n) \rangle dz \\ &= \sum \{ n\text{-point correlation functions} \} \end{aligned}$$

The meromorphic fields $v(z)$ are therefore generating functions for linear operators that act on $\mathcal{H}$.

Representation theory in conformal field theory

The subspaces V and $\bar{V}$ are a kind of algebra of fields called a vertex operator algebra and the space of states $\mathcal{H}$ is a representation of $V \otimes \bar{V}$.

Vertex operator algebras have two distinguished elements

- 1 The vacuum state which corresponds to the identity field/operator $\sum_{m \in \mathbb{Z}} \mathbf{1} \delta_{m,0} z^{-m}$.
- 2 The Virasoro element which corresponds to the generating function of infinitesimal conformal transformations $T(z) = \sum_{m \in \mathbb{Z}} L_m z^{-m-2}$, that is, its coefficients span the Virasoro algebra

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

The Virasoro commutation relations can also be compactly expressed by the generating functions

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg.}$$

Constructing conformal field theories from representation theory

Usually one is not handed a “solved” conformal field theory, that is, a space of states and its correlation functions. Instead all this structure needs to be constructed.

In this setting we

- 1 choose a vertex operator algebra V (assume $V \cong \bar{V}$),
- 2 classify the representation theory of V ,
- 3 determine the fusion products of V representations (Fusion encodes correlation functions at the level of representations. It is similar to the tensor product of Lie algebra representations.),
- 4 glue V -representations R_i together to form a space of states

$$\mathcal{H} = \sum_{i,j} R_i \otimes \bar{R}_j.$$

The Virasoro case

Today we will focus on representation theory in the pure Virasoro case, that is, when all fields in V are just polynomials of derivatives of $T(z)$.

Def: Virasoro Verma modules $M(h)$

Generated by a highest weight vector u_h , such that

$$L_0 u_h = h u_h, \quad L_n u_h = 0, \quad n \geq 1.$$

Basis consists of all vectors

$$L_{-n_1} \cdots L_{-n_k} u_h, \quad k \geq 0, \quad n_1 \geq n_2 \geq \cdots \geq n_k \geq 1.$$

The Virasoro commutation relations then imply that each above basis vector is an L_0 -eigenvector with eigenvalue $h + \sum_i n_i$.

The “minimal model” vertex operator algebras

We fix $c = 1 - 6\frac{(p-p')^2}{pp'}$, $p, p' \geq 2$ and coprime.

Def: The minimal model vertex operator algebras

As a Virasoro representation V is the irreducible quotient of $M(0)$

The subspaces by which one needs to quotient are generated by two singular vectors. Singular vectors are those which satisfy

$$L_0\chi = (h+m)\chi, \quad m \geq 1, \quad L_n\chi = 0, \quad n \geq 1.$$

Prop

The first singular vector is $L_{-1}u_0$. The second is very complicated. Its L_0 -eigenvalue is $(p-1)(p'-1)$.

The “minimal model” vertex operator algebras

Prop

V is spanned by the vectors $L_{-n_1} \cdots L_{-n_k} u_0$, $n_1 \geq \cdots \geq n_k \geq 2$.

The fields (generating functions) corresponding to these vectors are

$$L_{-n} u_0 \leftrightarrow \frac{\partial^{n-2} T(z)}{(n-2)!}, \quad L_{-n_1} \cdots L_{-n_k} u_0 \leftrightarrow: \frac{\partial^{n_1-2} T(z)}{(n_1-2)!} \cdots \frac{\partial^{n_k-2} T(z)}{(n_k-2)!} :.$$

Since the singular vector of eigenvalue $(p-1)(p'-1)$ has been set to 0, it must correspond to a field that acts trivially in any representation of V .

This is a very strong restriction and completely determines all V -representations!

Example: The Yang-Lee model, $p = 2, p' = 5$

For the Yang-Lee model $c = -\frac{22}{5}$ and the singular vector is given by

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})u_0 \leftrightarrow \chi(z) = :T(z)^2: - \frac{3}{10}\partial^2 T(z).$$

$\chi(z)$ must act trivially in any V -representation, so let's evaluate the z^{-4} coefficient of $\chi(z)$ on $u_h \in M(h)$:

$$\chi_0 = \oint \chi(z) z^3 \frac{dz}{2\pi i} = \sum_{m \leq -2} L_m L_{-m} + \sum_{m \leq -1} L_{-m} L_m - \frac{9}{5} L_0.$$

$$\chi_0 u_h = (L_1 L_{-1} + L_0^2 - \frac{9}{5} L_0) u_h = (h^2 + \frac{1}{5} h) u_h.$$

Thus, the action of $\chi(z)$ does not vanish if $h \neq 0, -\frac{1}{5}$.

With a bit more work one can show that there are only 2 irreducible V -representations (the irreducible quotients of $M(0)$ and $M(-\frac{1}{5})$) and that all representations are completely reducible.

The “minimal model” vertex operator algebras

As we just saw, singular vectors in V determine the representation theory. Also singular vectors in other representations are important for computing fusion products.

So how can we find singular vectors? The singular vector equation is intractable

$$L_1\chi = 0, \quad L_2\chi = 0$$
$$\chi = \sum_{\substack{n_1+\dots+n_k=(p-1)(p'-1) \\ k \geq 1 \\ n_i \geq n_{i+1} \geq 1}} A_{n_1, \dots, n_k} L_{-n_1} \cdots L_{-n_k} u_0.$$

The solution is given by so-called free field realisations and symmetric polynomials.

The free field realisation

Instead of considering the Virasoro algebra spanned by the L_m , we embed the Virasoro algebra into the larger but simpler algebra $\widehat{\mathfrak{gl}}(1)$. A basis of generators of $\widehat{\mathfrak{gl}}(1)$ is given by a_m , $m \in \mathbb{Z}$ with commutation relations

$$[a_m, a_n] = m\delta_{m+n,0}.$$

Or in terms of the generating function $a(z) = \sum_{m \in \mathbb{Z}} a_m z^{-m-1}$

$$a(z)a(w) = \frac{1}{(z-w)^2} + \text{reg.}$$

Def: $\widehat{\mathfrak{gl}}(1)$ Verma modules F_p

Generated by v_p , $a_m v_p = 0$, $m \geq 1$, $a_0 v_p = p v_p$.

Basis given by all $a_{-n_1} \cdots a_{-n_k} v_p$, $k \geq 0$, $n_1 \geq n_2 \geq \cdots \geq n_k \geq 1$.

Since the a_{-m} , $m \geq 1$ mutually commute, $F_p \cong \mathbb{C}[a_{-1}, a_{-2}, \dots] v_p$.

The free field realisation

So we are now considering the vertex operator algebra $\mathcal{F}$ that is generated by $a(z)$: its fields are polynomials of derivatives of $a(z)$.

Prop

As a $\widehat{\mathfrak{gl}}(1)$ -representation $\mathcal{F} \cong F_0$ and the fields corresponding to the elements of F_0 are

$$a_{-n_1} \cdots a_{-n_k} v_0 \leftrightarrow: \frac{\partial^{n_1-1} a(z)}{(n_1-1)!} \cdots \frac{\partial^{n_k-1} a(z)}{(n_k-1)!} :.$$

The generating function $T(z)$ and its coefficients are

$$T(z) = \frac{1}{2} : a(z)^2 : + \frac{p' - p}{\sqrt{2pp'}} \partial a(z), \quad L_0 = \frac{1}{2} a_0^2 + \frac{p - p'}{\sqrt{2pp'}} a_0 + \sum_{n \geq 1} a_{-n} a_n,$$
$$L_m = \frac{1}{2} \sum_{n \in \mathbb{Z}} a_n a_{m-n} + \frac{p - p'}{\sqrt{2pp'}} (m+1) a_m.$$

The solution in terms of symmetric polynomials

In $\mathcal{F}$ the singular vector $L_{-1}u_0$ is null, but the other, χ , is not. Does there exist a nice expression for χ in $\mathcal{F}$?

Yes! There are explicit formulae in terms of symmetric polynomials

$$\chi = \psi(\mathbf{P}_{[(p'-1)^{p-1}]}^{p/p'}(y))v_0,$$

where $\mathbf{P}_{[(p'-1)^{p-1}]}^{p/p'}(y)$ is a certain symmetric polynomial, called a Jack symmetric polynomial and ψ is an isomorphism of commutative rings from the ring of symmetric polynomials to $\mathbb{C}[a_{-1}, a_{-2}, \dots]$.

Symmetric polynomials

- One can consider symmetric polynomials in either a finite or infinite number of variables:
 - ▶ $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$
 - ▶ $\Lambda = \varprojlim_r \Lambda_r$ (also called symmetric functions) Λ_r can be obtained from Λ or $\Lambda_{r'}$, $r' > r$ by setting $z_i = 0$ for $i > r$.
- Both Λ and Λ_r are naturally graded by the degree of polynomials

$$\Lambda_r = \bigoplus_{m \geq 0} \Lambda_r^m, \quad \Lambda = \bigoplus_{m \geq 0} \Lambda^m.$$

Dimensions are

$$\dim \Lambda^m = \#\{\text{partitions of } m\}$$

$$\dim \Lambda_r^m = \#\{\text{partitions of } m \text{ of length at most } r\}.$$

Thus $\dim \Lambda^m = \dim \Lambda_r^m$ if $r \geq m$.

Symmetric polynomials

The rings Λ_r, Λ admit many bases and generators:

- The power sums $p_m = \sum_i z_i^m$ generate the ring of symmetric polynomials:

$$\Lambda_r = \mathbb{C}[p_1, \dots, p_r],$$

$$\Lambda = \mathbb{C}[p_1, p_2, \dots].$$

- The elementary symmetric polynomials

$$e_m = \sum_{1 \leq i_1 < \dots < i_m} z_{i_1} \cdots z_{i_m}$$

are also generators:

$$\Lambda_r = \mathbb{C}[e_1, \dots, e_r],$$

$$\Lambda = \mathbb{C}[e_1, e_2, \dots].$$

Symmetric polynomials

The monomial symmetric polynomials form a basis of Λ_r and Λ .

$$m_\lambda = \sum_{\sigma} z_{\sigma(1)}^{\lambda_1} z_{\sigma(2)}^{\lambda_2} \cdots,$$

where σ runs over all distinct permutations of λ . For example

- $m_{[2,1]}(z_1, z_2) = z_1^2 z_2 + z_1 z_2^2$
- $m_{[2,1]}(z_1, z_2, z_3) = z_1^2 z_2 + z_1 z_2^2 + z_1^2 z_3 + z_1 z_3^2 + z_2^2 z_3 + z_2 z_3^2$
- $m_{[s^r]}(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s$
- if $\ell(\lambda) > r$, then $m_\lambda = 0$ in Λ_r

Jack symmetric polynomials

Jack symmetric polynomials P_{λ}^{κ} are characterised by two properties:

- They are upper triangular in symmetric monomials with respect to the dominance ordering of partitions:

$$P_{\lambda}^{\kappa} = \sum_{\mu \leq \lambda} u_{\mu, \lambda}(\kappa) m_{\mu}, \quad u_{\lambda, \lambda}(\kappa) = 1,$$

where $\mu \leq \lambda$ if $|\mu| = |\lambda|$ and $\sum_{i=1}^n \mu_i \leq \sum_{i=1}^n \lambda_i$, $\forall n \geq 1$.

- Jack polynomials are mutually orthogonal with respect to the inner product

$$\langle f, g \rangle_r^{\kappa} = \int_{T^r} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j} \right)^{1/\kappa} f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}.$$

$$\langle P_{\lambda}^{\kappa}, P_{\mu}^{\kappa} \rangle_r^{\kappa} = 0 \quad \text{if } \lambda \neq \mu.$$

Jack symmetric polynomials

The dominance ordering of partitions is only partial, not total. Constructing the Jack polynomials by Gram-Schmidt orthogonalisation is therefore overdetermined. Yet it works (with a few restrictions on κ).

Important remarkable properties of Jack polynomials:

- The “norm square” $\langle P_\lambda^\kappa, P_\lambda^\kappa \rangle_r^\kappa$ is given by known combinatorial formulae.
- Jack polynomials satisfy the identity

$$\prod_{i,j \geq 0} (1 - y_i z_j)^{-1/\kappa} = \prod_{m \geq 1} \exp\left(\frac{1}{\kappa} \frac{p_m(y) p_m(z)}{m}\right) = \sum_{\lambda} b_{\lambda}(\kappa) P_{\lambda}^{\kappa}(y) P_{\lambda}^{\kappa}(z),$$

where the $b_{\lambda}(\kappa)$ are given by known combinatorial formulae.

- In Λ_r

$$P_{[s^r]}^{\kappa}(z_1, \dots, z_r) = m_{[s^r]}(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s, \quad \forall \kappa.$$

The strategy for finding singular vector formulae

The $\widehat{\mathfrak{gl}}(1)$ generators $a_{-n}, n \geq 1$ commute. Thus,
 $U(\widehat{\mathfrak{gl}}(1)_-) = \mathbb{C}[a_{-1}, a_{-2}, \dots]$.

Power sums generate Λ . Thus, $\Lambda = \mathbb{C}[p_1, p_2, \dots]$.

The secret to finding singular vector formulae lies in identifying these two polynomial rings appropriately.

Back to the free field theory

We consider so-called vertex operators

$$V_q(z) = e^{q\hat{a}} z^{qa_0} \prod_{m \geq 1} \exp\left(q \frac{a_{-m}}{m} z^m\right) \exp\left(-q \frac{a_m}{m} z^{-m}\right),$$

where $[a_m, \hat{a}] = \delta_{m,0}$. The operator $e^{q\hat{a}}$ shifts a_0 by q and one defines $e^{q\hat{a}} v_{q'} = v_{q+q'}$. Thus,

$$V_q(z) F_{q'} \rightarrow F_{q+q'}[[z, z^{-1}]] z^{qq'}.$$

The commutation relations of the L_m with $V_q(z)$ are

$$[L_m, V_q(z)] = \partial z^{m+1} V_q(z) + (h_q - 1)(m+1) V_q(z),$$

where $h_q = \frac{1}{2}q^2 + \frac{q}{2} \frac{p-p'}{\sqrt{2pp'}}$.

Virasoro homomorphisms from vertex operators

The commutator $[L_m, V_q(z)]$ is a total derivative when $h_q = 1$. There are two such q , $q_+ = \sqrt{\frac{2p'}{p}}$ and $q_- = -\sqrt{\frac{2p'}{p}}$. Then

$$Q_{\pm} = \oint_0 V_{q_{\pm}}(z)dz \quad \Rightarrow \quad [L_m, Q_{\pm}] = \oint \partial z^{m+1} V_{q_{\pm}}(z)dz = 0,$$

i.e., $Q_{\pm}$ is a Virasoro homomorphism.

This yields formulae for singular vectors! If $Q_{\pm}F_{q'} \rightarrow F_{q_{\pm}+q'}$ is well defined ($q_{\pm}q' \in \mathbb{Z}$) and $Q_{\pm}v_{q'} \neq 0$, then $Q_{\pm}v_{q'}$ is a Virasoro singular vector in $F_{q_{\pm}+q'}$.

Vertex operators and symmetric polynomials

$Q_{\pm}$ alone will not give us the singular vector we need. Instead consider products of vertex operators

$$\begin{aligned} Q_{\pm}^{[k]} &= \oint V_{q_{\pm}}(z_1) \cdots V_{q_{\pm}}(z_k) dz_1 \cdots dz_k \\ &= e^{kq_{\pm}\hat{a}} \oint \prod_{i \neq j} (z_i - z_j)^{q_{\pm}^2/2} \prod_{i=1}^k z_i^{q_{\pm}a_0} \\ &\quad \times \prod_{m \geq 1} \exp \left(q_{\pm} \frac{a_{-m} p_m(z)}{m} \right) \exp \left(-q_{\pm} \frac{a_m p_m(z^{-1})}{m} \right) dz_1 \cdots dz_k, \end{aligned}$$

where the $p_m(z)$ are the power sums.

As with $Q_{\pm}$, the $Q_{\pm}^{[k]}$ define Virasoro homomorphisms whenever the integration contour can be closed.

Evaluating the vertex operators

For $q' = (1-p)\sqrt{\frac{2p'}{p}}$ we have

$$\begin{aligned} Q_+^{[p-1]} v_{q'} &= e^{(p-1)q+\hat{a}} \oint \prod_{i \neq j} (z_i - z_j)^{p'/p} \prod_{i=1}^{p-1} z_i^{\sqrt{\frac{2p'}{p}} a_0} \\ &\quad \times \prod_{m \geq 1} \exp \left(\sqrt{\frac{2p'}{p}} \frac{a_{-m} \mathbf{p}_m(z)}{m} \right) v_{q'} dz_1 \cdots dz_{p-1}, \\ &= \oint \prod_{i \neq j} \left(1 - \frac{z_i}{z_j} \right)^{p'/p} \prod_{i=1}^{p-1} z_i^{1-p'} \\ &\quad \times \prod_{m \geq 1} \exp \left(\sqrt{\frac{2p'}{p}} \frac{a_{-m} \mathbf{p}_m(z)}{m} \right) v_0 \frac{dz_1 \cdots dz_{p-1}}{z_1 \cdots z_{p-1}} \end{aligned}$$

This looks like the inner product of Jack polynomials!

Evaluating the vertex operators

Let ψ be the ring isomorphism

$\psi : \Lambda \rightarrow \mathbb{C}[a_{-1}, a_{-2}, \dots], \mathbf{p}_m(y) \mapsto \sqrt{\frac{2p}{p'}} a_{-m}$, then

$$\begin{aligned} Q_+^{[p-1]} v_{q'} &= \oint \prod_{i \neq j} \left(1 - \frac{z_i}{z_j} \right)^{p'/p p-1} \prod_{i=1}^{p-1} z_i^{1-p'} \\ &\quad \times \psi \left(\prod_{m \geq 1} \exp \left(\frac{p'}{p} \frac{\mathbf{p}_m(y) \mathbf{p}_m(z)}{m} \right) \right) v_0 \frac{dz_1 \cdots dz_{p-1}}{z_1 \cdots z_{p-1}} \\ &= \sum_{\lambda} \left\langle \mathbf{P}_{[(p'-1)^{p-1}]}^{p/p'}(z), b_{\lambda} \left(\frac{p}{p'} \right) \mathbf{P}_{\lambda}^{p/p'}(z) \right\rangle \psi \left(\mathbf{P}_{\lambda}^{p/p'}(y) \right) v_0 \\ &= b_{[(p'-1)^{p-1}]} \left(\frac{p}{p'} \right) \psi \left(\mathbf{P}_{[(p'-1)^{p-1}]}^{p/p'}(y) \right) v_0. \end{aligned}$$

Evaluating the vertex operators

The singular vector

$$Q_+^{[p-1]} v_{q'} = \text{const} \cdot \psi \left(P_{[(p'-1)^{p-1}]}^{p/p'}(y) \right) v_0$$

can be converted into its corresponding field and the action of that field can again be evaluated using symmetric polynomial data to easily classify the representation theory of the minimal models.

Free field realisations are a very generic construction in conformal field theory and all of them appear to have interesting connections to symmetric polynomials.

The End

Thank you for your attention!